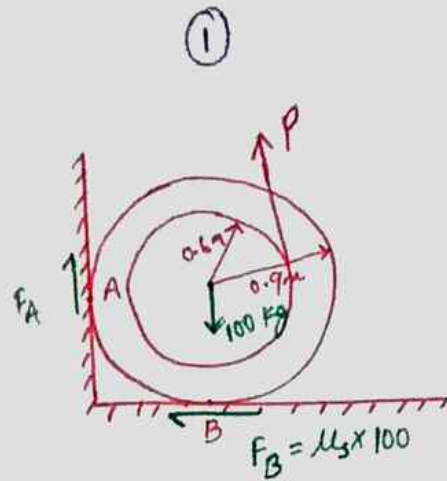


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Answer



Given,

weight of spool (w) = 100 kg

Coefficient of friction (μ_s) = 0.3

Taking moment about A,

$$\sum M_A = 0 \quad NB = W$$

$$\Rightarrow -P \times (0.9 + 0.6) + F_B \times 0.9 + 100 \times 0.9 = 0$$

$$\Rightarrow -P \times 1.5 + \mu_s \times 100 \times 0.9 + 90 = 0$$

$$\Rightarrow -1.5P + 0.3 \times 100 \times 0.9 + 90 = 0$$

$$\Rightarrow 1.5P = 117$$

$$\Rightarrow P = \frac{117}{1.5}$$

$$P = 78 \text{ kg}$$

Maximum force that can be applied is 78 kg.

Friction force is given as

Friction force = (coefficient of friction) $\times$ (normal force)

Thus by taking moment about point A the load P can be found.

Likes: 0

Dislikes: 0
